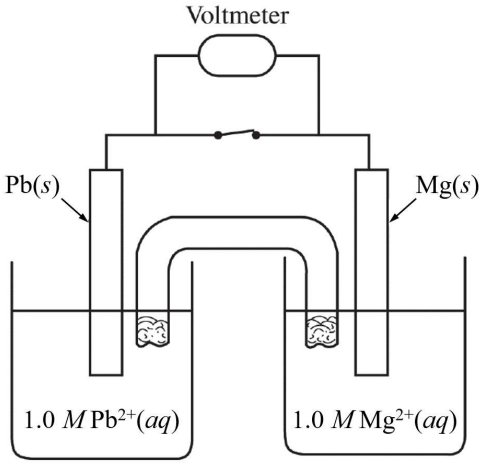
Topics 9.8 – 9.11: MCQ Practice



Half-Reaction	$E°$ (V)
$Pb^{2+}(aq) + 2\,e^- \rightarrow Pb(s)$	–0.13
$Mg^{2+}(aq) + 2\,e^- \rightarrow Mg(s)$	–2.37

A student sets up a standard galvanic cell with lead (Pb) and magnesium (Mg), as shown in the diagram above. Standard reduction potentials for the half-reactions associated with the galvanic cell are given in the table above.

1. The student fills the salt bridge with a solution of potassium nitrate, $KNO_3(aq)$. Which of the following is the correct description of the movement of potassium ions in the salt bridge as the cell operates?

   (A) Potassium ions flow toward the Pb half-cell compartment, in which the oxidation half-reaction occurs.

   (B) Potassium ions flow toward the Pb half-cell compartment, in which the reduction half-reaction occurs.

   (C) Potassium ions flow toward the Mg half-cell compartment, in which the oxidation half-reaction occurs.

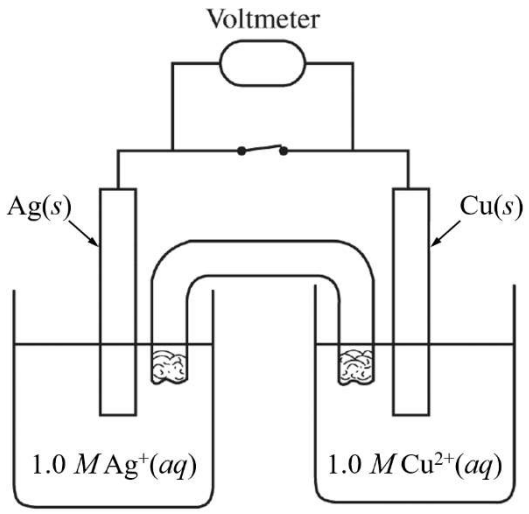
   (D) Potassium ions flow toward the Mg half-cell compartment, in which the reduction half-reaction occurs.

Half-Reaction	$E°$ (V)
$Ag^{+}(aq) + e^- \rightarrow Ag(s)$	+0.80
$Cr^{3+}(aq) + 3\,e^- \rightarrow Cr(s)$	–0.74

2. The standard reduction potentials of the half-reactions associated with a galvanic cell are listed in the table above. Which of the following shows the correct values for the cell potential ($E°$) and the Gibbs free energy change ($\Delta G°$) for the overall reaction that occurs in this standard galvanic cell?

	$E°$ (V)	$\Delta G°$ (kJ/mol$_{rxn}$)
(A)	+1.54	–149
(B)	+1.54	–446
(C)	+3.14	–303
(D)	+3.14	–909

Questions 3 – 4 refer to the following information.



A student sets up a standard galvanic cell with silver (Ag) and copper (Cu), as shown in the diagram above. Standard reduction potentials for the half-reactions associated with the galvanic cell are given in the table below.

Half-Reaction	$E°$ (V)
$Ag^+(aq) + e^- \rightarrow Ag(s)$	+ 0.80
$Cu^{2+}(aq) + 2\,e^- \rightarrow Cu(s)$	+ 0.34

3. Which of the following statements identifies the electrode that decreases in mass as the cell operates, and correctly explains why?

   (A) The Ag electrode, because the standard reduction potential of Ag is larger than that of Cu

   (B) The Ag electrode, because 2 moles of Ag are consumed for every 1 mole of Cu in this reaction

   (C) The Cu electrode, because atoms on the Cu(s) electrode are oxidized to form $Cu^{2+}(aq)$ ions

   (D) The Cu electrode, because Cu has a smaller molar mass than Ag has

4. What is the value of $E°$ for this galvanic cell?

   (A) 0.46 V

   (B) 1.14 V

   (C) 1.26 V

   (D) 1.94 V

Questions 5 – 6 refer to the following information.

The standard reduction potentials of some half-reactions are given in the table below.

Half-Reaction	$E°$ (V)
$Al^{3+}(aq) + 3\,e^- \rightarrow Al(s)$	–1.66
$Zn^{2+}(aq) + 2\,e^- \rightarrow Zn(s)$	–0.76
$Cu^{2+}(aq) + 2\,e^- \rightarrow Cu(s)$	+0.34
$Rh^{3+}(aq) + 3\,e^- \rightarrow Rh(s)$	+0.80

5. Of the following equations, which represents the reaction that has the greatest thermodynamic favorability?

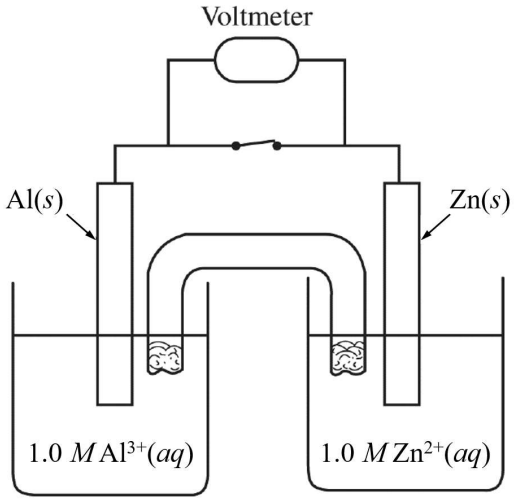
(A) $Al^{3+}(aq) + Rh(s) \rightarrow Al(s) + Rh^{3+}(aq)$

(B) $Rh^{3+}(aq) + Al(s) \rightarrow Rh(s) + Al^{3+}(aq)$

(C) $3\,Cu^{2+}(aq) + 2\,Rh(s) \rightarrow 3\,Cu(s) + 2\,Rh^{3+}(aq)$

(D) $2\,Rh^{3+}(aq) + 3\,Cu(s) \rightarrow 2\,Rh(s) + 3\,Cu^{2+}(aq)$

A student sets up a standard galvanic cell with aluminum (Al) and zinc (Zn), as shown in the diagram below.



6. The value of $\Delta G°$ for the reaction that occurs in this cell is closest to which of the following?

(A) –520 kJ/mol$_{rxn}$

(B) –260 kJ/mol$_{rxn}$

(C) –170 kJ/mol$_{rxn}$

(D) –87 kJ/mol$_{rxn}$

$$2\ Cu^{2+}(aq)\ +\ 2\ H_2O(l)\ \rightarrow\ 2\ Cu(s)\ +\ O_2(g)\ +\ 4\ H^{+}(aq) \qquad E^\circ = -0.89\ V$$

7. Which of the following is true for the reaction represented by the equation shown above?

(A) $\Delta G^\circ < 0$ and $K_{eq} < 1$

(B) $\Delta G^\circ < 0$ and $K_{eq} > 1$

(C) $\Delta G^\circ > 0$ and $K_{eq} < 1$

(D) $\Delta G^\circ > 0$ and $K_{eq} > 1$

$$2\ I^{-}(aq)\ +\ 2\ H_2O(l)\ \rightarrow\ I_2(aq)\ +\ H_2(g)\ +\ 2\ OH^{-}(aq) \qquad \Delta G^\circ = +264\ kJ/mol_{rxn}$$

8. The electrolysis of an aqueous solution of potassium iodide, KI(aq) results in a chemical reaction that produces iodine, $I_2(aq)$, according to the net ionic equation shown above. What is the approximate minimum voltage that must be supplied to a 1.0 M solution of KI(aq) to produce $I_2(aq)$ according to reaction shown above?

(A) 0.0027 V

(B) 0.70 V

(C) 1.4 V

(D) 2.7 V

Questions 9 – 10 refer to the following information.

A student sets up a standard galvanic cell with aluminum (Al) and silver (Ag). The cell potential is measured to be 2.46 V, which corresponds to the value of the standard cell potential ($E°$) for the reaction represented by the following equation.

$$Al(s) + 3\ Ag^{+}(aq) \rightarrow Al^{3+}(aq) + Ag(s) \quad E° = +2.46\ V$$

9. In a second trial of this experiment, a student sets up an identical standard cell to the one described above. Then a few drops of 2.0 *M* NaCl(*aq*) are added to the half-cell containing $Ag^{+}(aq)$, resulting in the formation of a precipitate of AgCl(*s*). Which of the following choices correctly compares the measured value of the cell potential (E) in Trial 2 to the standard cell potential ($E°$) observed in Trial 1 and provides the correct justification in terms of the Nerst equation?

	Measured Cell Potential (E) in Trial 2	Justification in terms of the Nernst Equation
(A)	E is less than 2.46 V	Q is less than 1
(B)	E is less than 2.46 V	Q is greater than 1
(C)	E is greater than 2.46 V	Q is less than 1
(D)	E is greater than 2.46 V	Q is greater than 1

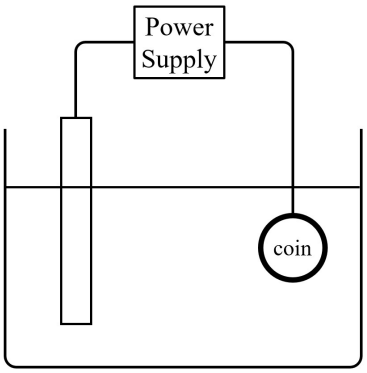
10. In a third trial of this experiment, a student sets up a galvanic cell made from aluminum (Al) and silver (Ag). The cell potential is measured to be 2.50 V. Which of the following is most likely to account for the observation that the measured cell potential (E) in Trial 3 is greater than the standard cell potential ($E°$) observed in Trial 1?

(A) The Ag(*s*) electrode has a greater mass than the Ag(*s*) electrode used in Trial 1.

(B) The Al(*s*) electrode has a greater mass than the Al(*s*) electrode used in Trial 1.

(C) The concentration of $Ag^{+}(aq)$ in the Ag half-cell has been changed from 1.0 *M* to 0.010 *M*.

(D) The concentration of $Al^{3+}(aq)$ in the Al half-cell has been changed from 1.0 *M* to 0.010 *M*.



$$Au(CN)_4^-(aq) + 3\,e^- \rightarrow Au(s) + 4\,CN^-(aq)$$

11. An electroplating experiment is performed in which a bronze coin is immersed in a solution containing $Au(CN)_4^-(aq)$. Electricity is passed through the solution, and a thin layer of solid gold, $Au(s)$, is deposited onto the surface of the coin according to the reaction represented by the equation shown above. If 4.00 A of current is applied to the cell, the time required to plate 2.00 g of $Au(s)$ onto the surface of the coin is closest to which of the following?

    (A) 81.6 s

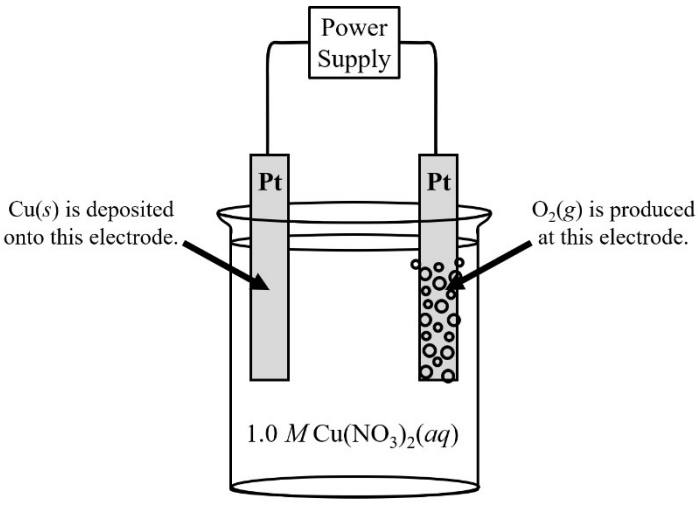
    (B) 245 s

    (C) 735 s

    (D) 11,800 s

12. An external direct-current power supply is connected to two platinum electrodes immersed in a beaker containing 1.0 *M* $Cu(NO_3)_2(aq)$, as shown in the diagram at right. As the cell operates, $Cu(s)$ is deposited onto one electrode, and $O_2(g)$ is produced at the other electrode.

    An electric current of 2.50 A passes through the cell for 35.0 minutes. The mass of $Cu(s)$ that is deposited on the electrode is closest to which of the following?

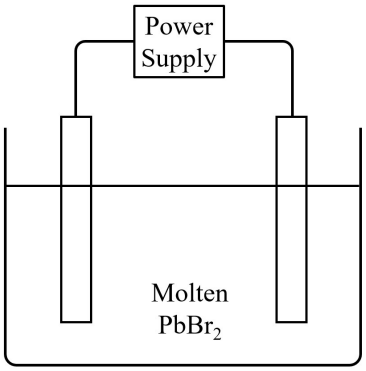


    (A) 0.277 g Cu

    (B) 1.73 g Cu

    (C) 3.46 g Cu

    (D) 6.92 g Cu



$$PbBr_2(l) \rightarrow Pb(l) + Br_2(g)$$

13. An experiment is performed in which molten lead bromide, $PbBr_2(l)$ is electrolyzed using an external power supply and inert electrodes, as represented in the diagram above. Lead metal and bromine gas are produced according to the equation shown above. If 5.0 g of lead metal is produced in a period of 1800 seconds, the average current flowing through the cell in this experiment is closest to which of the following?

(A) 0.39 C/s

(B) 0.65 C/s

(C) 1.3 C/s

(D) 2.6 C/s